Predicting streetcar bunching risk-events from risk-patterns

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An ML approach to congestion on the TTC

Domain introduction

- In March 2025, the Toronto Transit Commission (TTC) launched a "Bunching and Gapping Pilot" project, as an attempt to "*...improve performance centred on bunching (vehicles too close together) and gapping (vehicles too far apart) that disrupt service regularity, reduce capacity, and lower customer satisfaction."*"
- A pair of streetcars is considered to be bunched, if they arrive in quick succession of each other. In this project, I consider a pair of streetcars to be bunched if they arrive 50% sooner than their scheduled headway.

This is a (modern) TTC streetcar.

- One major issue was the high level of immediate supervision required to handle bunching issues. It was suggested that this level of supervision would not scale realistically to the entire system.

AI / Machine Learning to the rescue?

Can one develop machine learning prediction tools to help with route supervision?

- How could ML prediction tools fit into a reasonable route supervision framework?
- Can these ML prediction tools operate with data that can be realistically gathered for time of deployment?

Potential route supervisory framework

Data input

TTC command receives and logs real-time AVL/GTFS data for each streetcar arrival at each stop.

Response/solution

Either by direct supervisor intervention, or by another ML tool, provide an appropriate response to a risk flag (for example: a suggested holding time...)

Risk flagging

ML prediction tool: make a prediction if a bunching incident may occur in the next stop, or at stops in a vicinity of adjacent stops.

Project specification

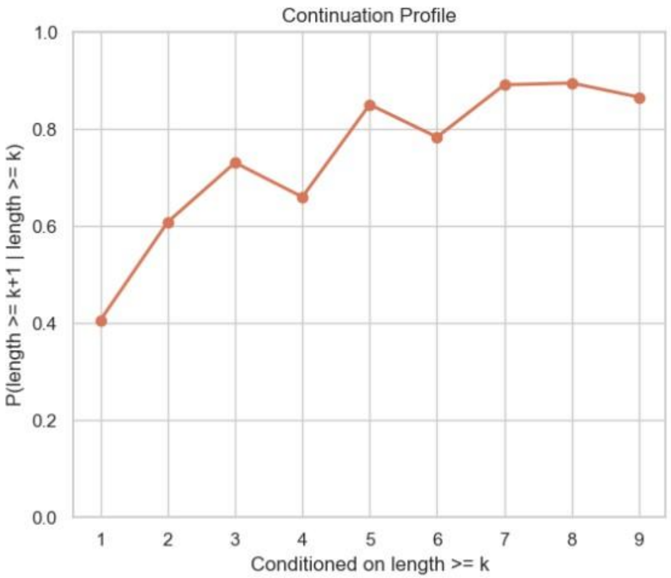
- The aim of this project is to lay the foundations for building the risk flagging component of the supervisory framework.
- In the 2025 TTC Bunching and Gapping Pilot, two specific streetcar lines were targeted: 506 Carlton, and 512 St. Clair. In this project, we focus on the 506 Carlton.
- This project provides predictive models for a variety of different risk classification targets.

Data sources

- The majority of the data used in this project is from TransSee (https://www.transsee.ca)
- TransSee aggregates live vehicle AVL/GTFS information, and organizes the data into a convenient database.
    - Used BeautifulSoup to scrape this data. Resulting in around 5.3 million data points.
- The model is also trained on various external factors. These include active road permits, traffic signals, pedestrian crossings in the vicinity of the 506 line. Also included are certain weather features.
- External data is sourced from Open Data Portal Toronto and Environment Canada.

Data insights

- A key insight gathered from the raw stop-level streetcar arrival dataset is that if a pair of streetcars have already been bunched for a certain number of stops, then there is a moderately high probability that they will arrive bunched again at future stops.
- To confirm this, I needed to attach stop location data to the raw stop-level dataset, and order the stop locations by stop precedence (depending on direction of travel).
- With the stop location data attached to these streetcar arrival entries, we can have a better grasp on the sequential dynamics involved with bunching patterns.



From data insights to modelling idea

- In light of the data insights, the idea for the project naturally turned to the following:

Can we predict if a certain pattern of bunching incidents between a pair of streetcars will continue in the future to also include bunching incidents at future stops?

- To take this question seriously, we need to qualify what we mean by "pattern of bunching incidents", and in what way they can continue to include bunching incidents at future stops.
- This involves some heavy feature engineering, involving flagging groups of bunching incidents that are both sequential in stop order and in time.

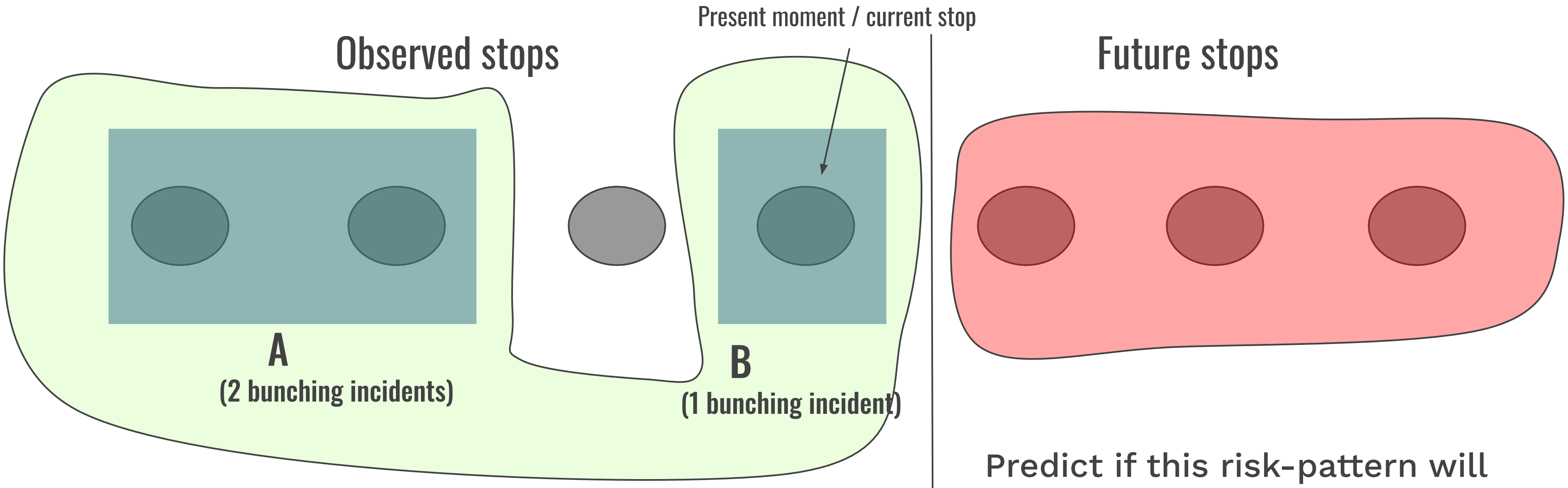
Risk-patterns and risk-events

- Risk-events are predefined for each model. Each type of risk-event is considered as an individual binary classification task.
- Risk-event prediction tasks are set by specifying integers **(b,s,n,m)**. For **(b,s,n,m)**, the associated prediction task is:

  "given a number of bunching incidents ≥ b, observed by a pair of streetcars in the last s stops, predict if the same pair of streetcars will encounter another number of bunching incidents ≥ n within the next m stops"

- A **risk-pattern** is a collection of bunching episodes that when taken together, satisfy the conditional premise of a risk-event prediction task
- A **risk-event** is a risk-pattern with a *true* positive outcome for the associated risk-event prediction task

Given 3 bunches in the last 4 stops.. Predict if a bunching incident will occur somewhere within the next 3 stops (illustrated)



Two continuous bunching episodes, A and B, grouped together gets flagged as a "risk-pattern"

Predict if this risk-pattern will extend to include also a bunching incident somewhere in these future stops.
(i.e. if this risk-pattern is a true "risk-event")

Features related to external factors

- In addition to certain lagged features assigned to each bunching episode (for example, the state of the bunched pair at the start of the episode), the model is also trained on various external factors.
- These include: active road permits, traffic signals, pedestrian crossings in the vicinity of the 506 line. Also included are certain weather features.
- External data is sourced from Open Data Portal Toronto and Environment Canada.

Model pipeline

Input: stop-level arrival data and external factors data

Extract bunching incidents and build dataframe of bunching episodes

Build "incidents" dataframe, marking episodes that are associated with "risk-patterns"

Incidents dataframe is fed into a binary classifier for each type of risk-event

Two layers of feature engineering

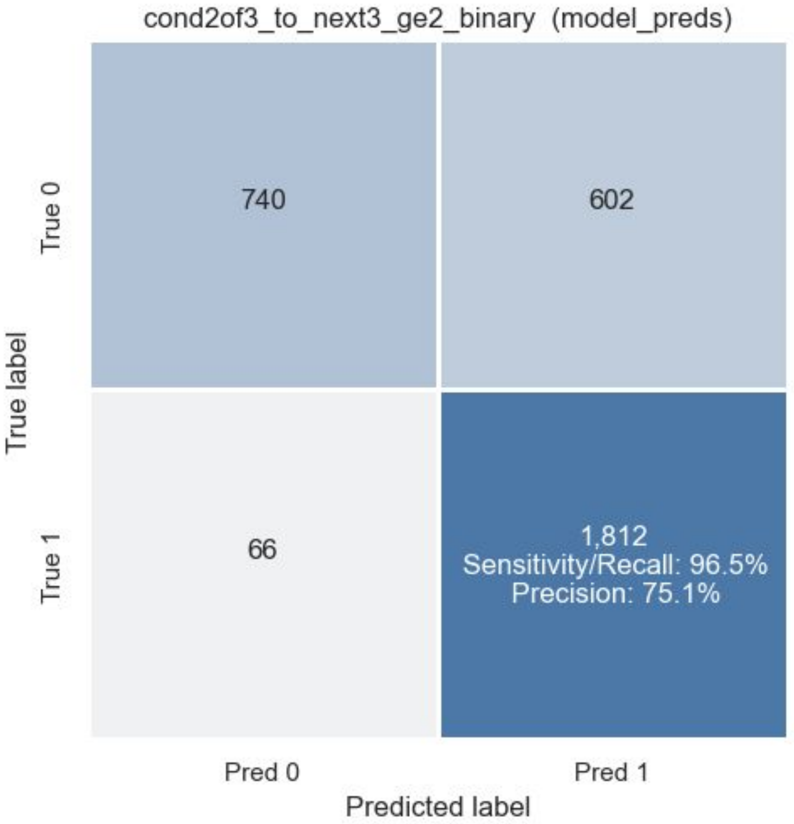
Time sensitive train/validation/test splits are required to prevent future data leakage.

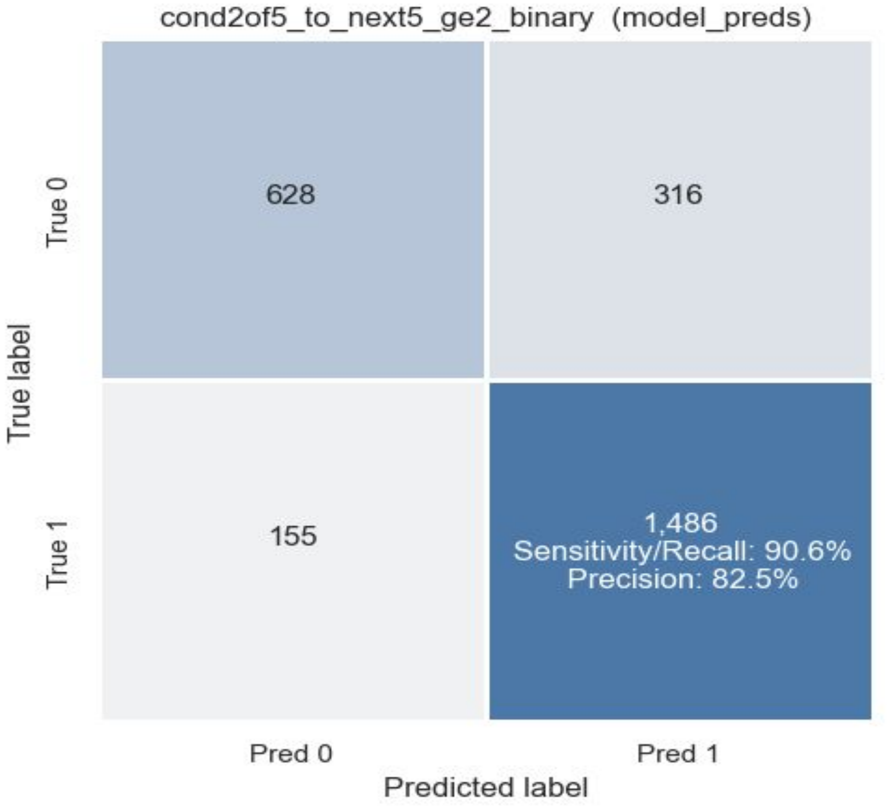
Classifiers are trained as XGBoost models, with hyperparameter and threshold tuning

Evaluating model performance: important metrics

- As our envisioned primary use-case for the prediction model is for risk-flagging, we pay closer attention to the metrics of recall, precision and false positive rate.
- We want to ensure a sufficiently high recall rate, but also make sure that the false positive rate is below an acceptable level.
- The probabilistic classifier is threshold-tuned according to a policy aligned with these goals.

Performance results for a sample of risk-events between a pair of streetcars

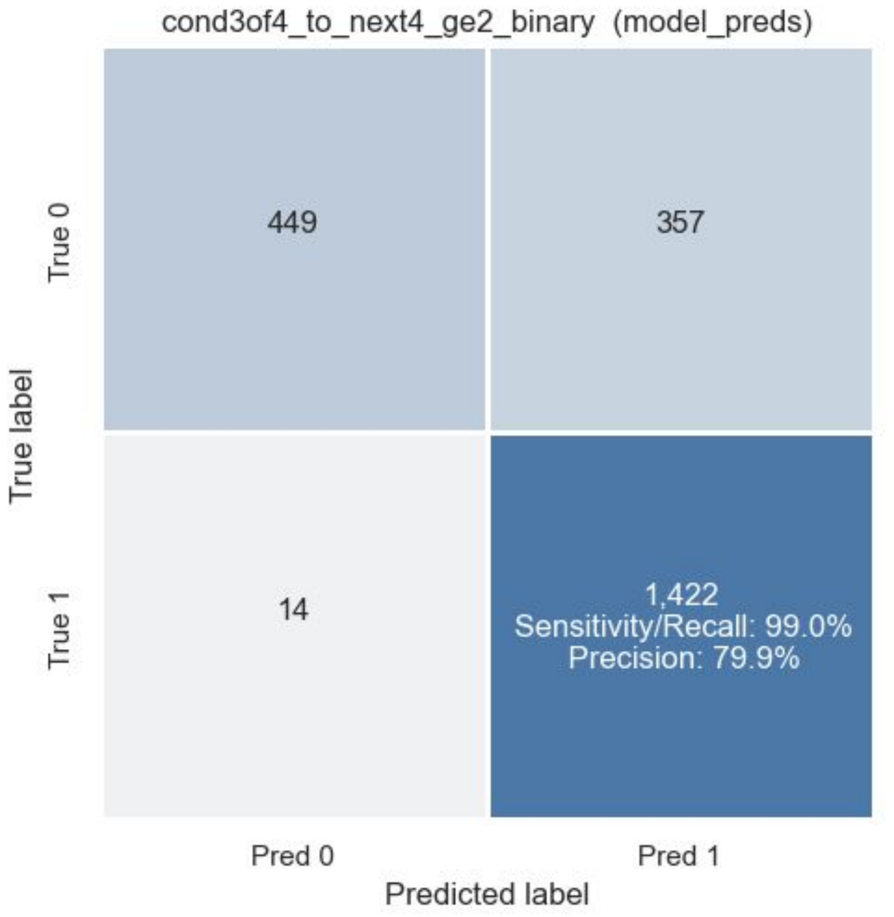


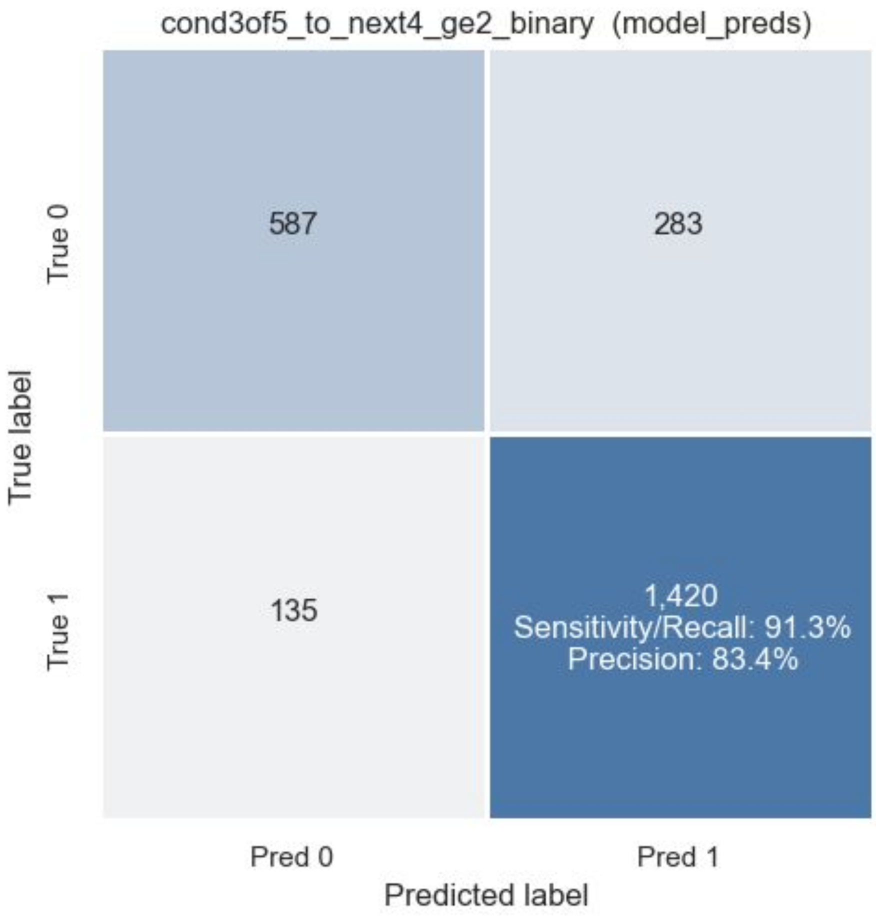


Event Type

Given 2 bunching events in the last 3 stops, will there be 2 more bunch events in the next 3 stops

Given 2 bunching events in the last 5 stops, will there be 2 more bunch events in the next 5 stops



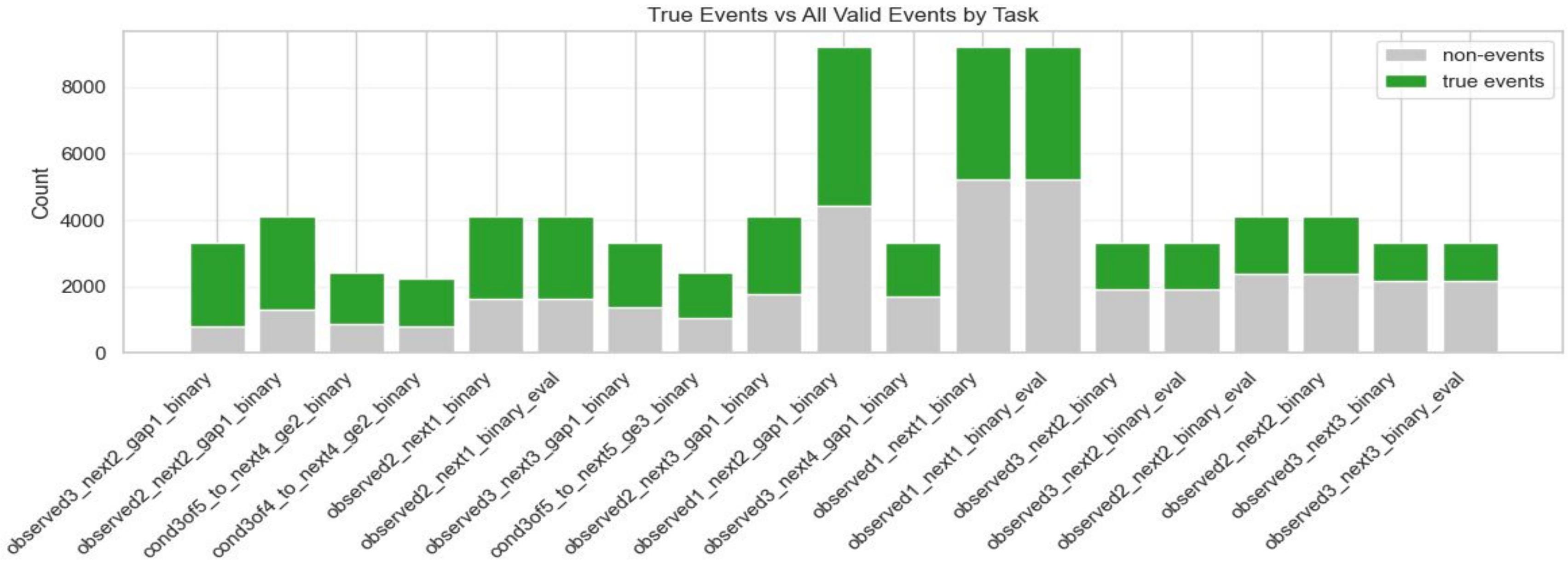


<u>Event Type</u>

Given 3 bunching events in the last 4 stops, will there be 2 more bunch events in the next 4 stops

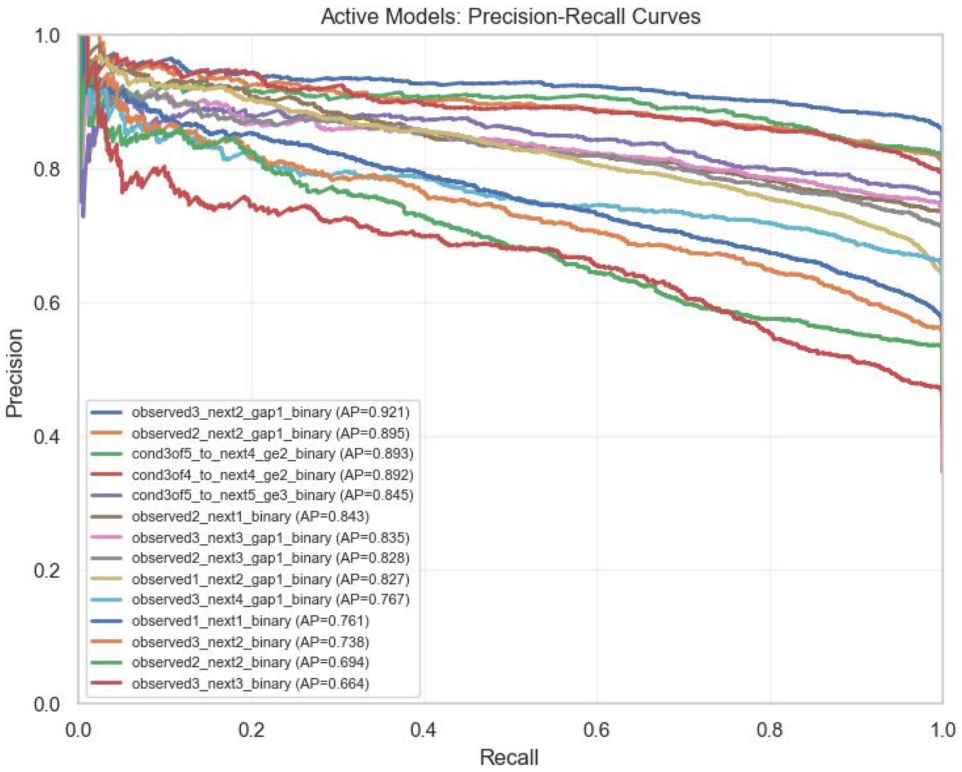
Given 3 bunching events in the last 5 stops, will there be 2 more bunch events in the next 4 stops

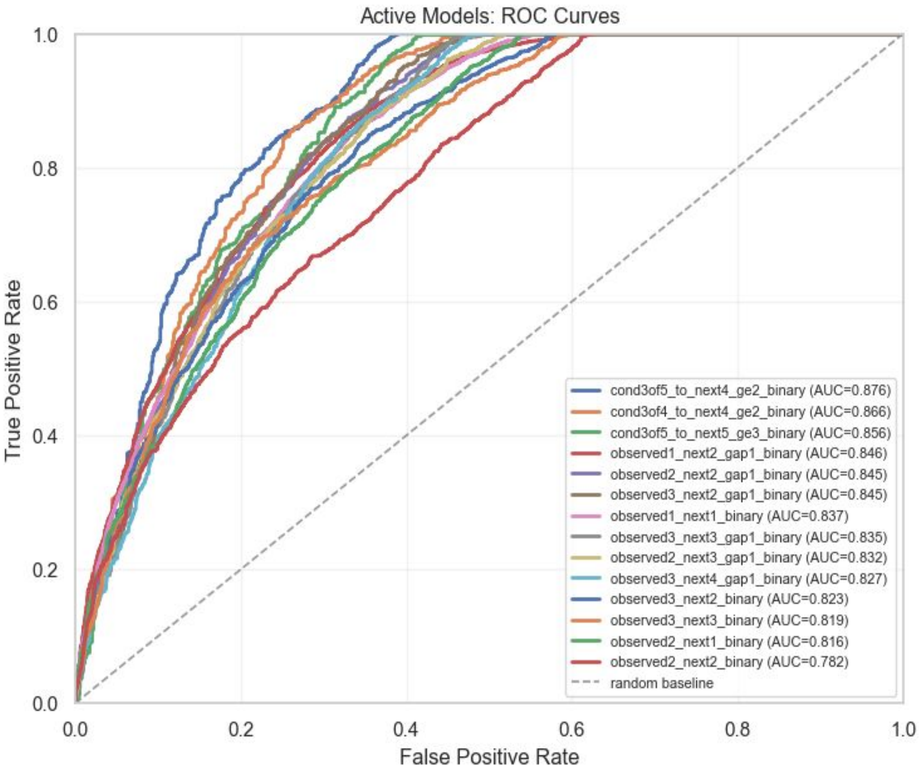
Risk-pattern v.s. Risk-event (conditional on valid pattern)



- For each type of risk-event, with a conditional prior risk-pattern, the green portion represents the portion of risk-patterns that extend to risk-events after the stop at which the risk-pattern was observed.
- Some are more skewed than others, nothing is too skewed.
- More balanced events will result in more balanced models without specific tuning.

Precision-Recall and ROC curves





Model performance scales with how ambitious the risk-pattern task is...

Limitations and next steps

- As already described, this model is only concerned with risk flagging. A full analysis on metric importance will ultimately depend on solution responses to such flags.
- For next steps, it would be interesting to build a solution-suggestion model for a given flag. One idea can be for such a model to run through various "what-if" scenarios and evaluate associated risks.
- Of course, these models should be trained for a variety of risk-tasks for streetcar lines other than the 506 Carlton.

Thank you!